0.1 Why the Response Remains of the Closed Form: Proposition, Bound, and Verification

0.1.1 Exact dynamics and the structure of the perturbations

About the active pivot line P, the exact roll dynamics during ground contact read (the pitch case is analogous)

$$J_P\,\dot{\omega} = M_{x,P}(t) + \Delta M_{\mathrm{GE},P}(t,\phi) - W(l_p + \lambda_{\mathrm{off}})\cos\phi + W z_{\mathrm{CoM}}\sin\phi, \tag{1}$$

with $M_{x,P} = M_x + f\,l_p$ and, conservatively, $l_{p,\phi} = 0.140$ m, $l_{p,\theta} = 0.110$ m. The ground-contact phase is a one-degree-of-freedom rigid rotation about the gear line, so every rotor height is an exact trigonometric function of the single angle ϕ; in the level configuration all heights coincide, yielding the exact identity

$$\Delta M_{\mathrm{GE},P} = \eta(\phi)\,\big(M_x + f l_p\big), \qquad \eta(\phi) \triangleq \bar{\gamma}(\phi) - 1, \tag{2}$$

where $\bar{\gamma}$ denotes the rotor-common thrust factor. The identity is *independent of the CoM position* — λ_{off} enters (1) only through the gravity arm — so ground effect biases the identified critical moment identically for every CoM configuration. The magnitude is evaluated with two models of increasing physics content. The per-rotor superposition of the single-rotor Cheeseman–Bennett result (which *neglects* rotor–rotor interference) gives, at the measured geometry ($h = 0.315$ m, $R = 0.127$ m), $\eta(0) = 1.03\%$ independent of l_p. The reported model adds the interference from first principles by the image method: each rotor is a three-dimensional potential source of strength $R^2 v_\infty/4$, the ground plane is enforced by mirror images, and only the vertical projection of each image-induced velocity reduces the downwash, giving the per-rotor sum

$$s_{\mathrm{rot},i} \;=\; \frac{R^2}{4}\sum_j \frac{z_i + z_j}{\big[d_{ij}^2 + (z_i + z_j)^2\big]^{3/2}}, \qquad \gamma_i = \big(1 - s_{\mathrm{rot},i}\big)^{-1}, \tag{3}$$

evaluated at each rotor's *own* tilted height $z_i = h\cos\phi + s_i\sin\phi$ — no averaging enters, and the attitude dependence of the model resides entirely in this term. The self term $j = i$ reduces exactly to Cheeseman–Bennett, so (3) contains the single-rotor result as the zero-neighbour limit and adds *no empirical constant of any kind*: the neighbour sum is fixed by the measured rotor geometry alone. Mapping the per-rotor gains to the pivot,

$$\Delta M_{\mathrm{GE}} = \sum_i b_i\,(\gamma_i - 1)\,T_i \;+\; \Big[\sum_i (\gamma_i - 1)\,T_i\Big] l_p, \tag{4}$$

with b_i the body-centre arms and the second term the parallel-axis transfer of the net vertical surplus. Under constant total thrust the result is affine in the commanded moment, $\Delta M_{\mathrm{GE}} = a(\phi) + b(\phi)M_{\mathrm{cmd}}$, with fitted coefficients $a/(fl_p)\,[\%] = 4.27 - 0.086\,\phi$ and $b\,[\%] = 4.28 - 0.060\,\phi$ (ϕ in degrees) — a factor ≈ 4 above the interference-free value in both channels, which is the quantitative statement that rotor–rotor interference is not negligible at this clearance. The affine decomposition, the anti-restoring sign of the height-asymmetry moment, and the channel structure below are identical in the two models — they follow from thrust-proportionality and smooth height dependence alone; only the magnitudes differ.

Everything that follows is deliberately insensitive to that magnitude. The absorption argument assumes only smoothness and thrust-proportionality, which any near-ground aerodynamic model shares, and the same is true of a purely geometric alternative: a contact-force lever displaced inboard of the kinematic pivot circle by a finite, compliant contact patch enters (1) in the same functional position as the a-channel. The static check of the identified thresholds indeed cannot separate the two — a 19.2 ± 1.8 mm inboard lever reproduces the measured deficit as well as the interference model does (residual RMS 82.3 versus 86.0 mN m), the discriminating moment-proportional channel being below the group-to-group scatter. This is a limitation of the attribution, not of the method: the bounds below hold for either cause, and both cancel antisymmetrically in the pivot-free deliverable.

0.1.2 Ideal configuration: the cosh family and its escape channel

Consider first the ideal level configuration: the vehicle rests at $\phi = 0$, so the onset — defined dynamically by zero net moment, $\omega(0) = \dot{\omega}(0) = 0$ — occurs *at* $\phi = 0$, and the operating point $(\tau, \phi, T) = (0, 0, T^*)$ makes the onset expansion and the textbook small-angle expansion one and the same. The tilt variable and the excursion coincide, $\delta\phi = \phi$.

Linearizing (1) about this point collects every perturbation into three absorbable channels plus a remainder: constants are absorbed by the onset anchoring; terms $\propto \dot{M}\tau$ (including the thrust-ramp channel of (2), whose coefficient η^* is rate-independent) are absorbed by the ramp-invariance calibration of K; terms $\propto \delta\phi$ (the gravity slope $W z_{\mathrm{CoM}}$ — here an *anti-restoring* gradient, i.e. a negative stiffness, which is what makes the response hyperbolic rather than oscillatory — and $\eta' M_P^*$) are absorbed by the calibrated C_2. What remains is the second-order Taylor remainder in Lagrange form,

$$\rho \;=\; \tfrac{1}{2}\, G''(\xi)\,\delta\phi^2 + \ldots, \qquad G''(\xi) = W(l_p + \lambda_{\mathrm{off}})\cos\xi - W z_{\mathrm{CoM}}\sin\xi, \tag{5}$$

with ξ between 0 and ϕ: the sine term enters at second order *with a negative sign* and is dropped conservatively, $|G''| \le W(l_p + \lambda_{\mathrm{off}})$. At the 5° excursion this bound evaluates to 19.2 mN m (roll) / 15.6 mN m (pitch).

Proposition 1 *Under (i) maintained pivot contact, (ii) gate-passing ramp execution, and (iii) any thrust-proportional, smooth ground-effect model, subtracting the nominal (onset-balanced) equation from* (1) *yields for the deviation* $e_\phi \triangleq \phi - \phi_{\mathrm{nom}}$

$$J_P\,\ddot e_\phi - W z_{\mathrm{CoM}}\, e_\phi = \rho(\tau), \qquad e_\phi(0) = \dot e_\phi(0) = 0, \tag{6}$$

where the state-dependent remainder is recast as an equivalent exogenous input, $\rho(\tau) \triangleq \rho(\delta\phi(\tau))$, and $W z_{\mathrm{CoM}} = J_P C_2^2$ is understood as the effective *anti-restoring gradient that the calibrated C_2 estimates. Every solution of the unforced problem with the onset conditions $\omega(0) = \dot\omega(0) = 0$ is exactly $\omega(\tau) = C_1(\cosh C_2\tau - 1)$ with $C_1 = K\dot M$ fixed — no shape parameter is free. Ground effect and the gravity nonlinearity therefore move the system* within *the two-parameter cosh family; the only escape channel is ρ.*

Impulse response and the propagated bound. The response of (6) to a unit moment impulse $\rho = \delta(\tau)$ follows from the momentum jump $J_P[\dot e_\phi(0^+) - \dot e_\phi(0^-)] = 1$ (the position cannot jump) and the unstable homogeneous solution: $e_\phi = \sinh(C_2\tau)/(J_P C_2)$, hence the rate impulse response $h_\omega(\tau) = \dot e_\phi = \cosh(C_2\tau)/J_P$ — consistent with $h_\omega(0) = 1/J_P$ from the jump condition. Superposition (Duhamel's integral [*ref*]) then gives the exact solution for arbitrary ρ and, normalizing like the empirical fit metric by the window-edge signal, the finite monotone envelope

$$\delta\omega(\tau) = \frac{1}{J_P}\int_0^\tau \cosh\big(C_2(\tau - s)\big)\rho(s)\,ds, \qquad \frac{|\delta\omega(\tau)|}{\omega_{\mathrm{nom}}(\tau_{\mathrm{end}})} \le \frac{C_2\,\rho_{\max}}{\dot M}\,\frac{\sinh(C_2\tau)}{\cosh(C_2\tau_{\mathrm{end}}) - 1}, \tag{7}$$

with $\rho_{\max} \triangleq \sup_{[0,\tau_{\mathrm{end}}]}|\rho|$, attaining $(C_2\rho_{\max}/\dot M)\coth(C_2\tau_{\mathrm{end}}/2)$ at the edge. Three points sharpen the bound. (a) *Kernel–forcing separation.* The onset conditions give the coherent forcing a sixth-order zero at the anchor ($\omega_{\mathrm{nom}} \sim \tau^2$, hence $\delta\phi \sim \tau^3$ and $\rho \sim \delta\phi^2 \sim \tau^6$, leading order for $C_2\tau \lesssim 1$), which is exactly where the unstable kernel amplifies most; conversely the forcing reaches $\rho_{\max}$ only at the window edge, where the kernel is unity. At the window midpoint the forcing is below 1% of its edge value. The quoted deviations are therefore *not* obtained from the uniform bound: the exact nominal remainder profile,

$$\rho(s) = \tfrac{1}{2}\, G_2 \left[\tfrac{C_1}{C_2}\big(\sinh C_2 s - C_2 s\big)\right]^2, \qquad G_2 = W(l_p + \lambda_{\mathrm{off}})\,(1 + 5\%_{\mathrm{GE}}), \tag{8}$$

is convolved with the kernel by direct numerical quadrature of (7) — no series truncation; the τ^6 law is only the small-argument limit, and the exact profile is in fact *more* edge-concentrated than the power law, its tail growing exponentially. This exact propagation tightens the uniform bound by a factor 25–96; a hand-derivable version splits the window at its midpoint, where the early half carries less than 1/128 of the forcing mass while the late-half kernel stays below $\cosh(C_2\tau_{\mathrm{end}}/2)$, squeezing the convolution between 1 and ≈2.2 times the plain integral of ρ. Since (8) enters *without* first removing its span-absorbable part, the result is doubly conservative. (b) *State dependence.* A priori the remainder is evaluated along the nominal trajectory; the small-gain closure with $L = \max|d\rho/d\phi| \approx 0.95$ N m/rad $\ll W z_{\mathrm{CoM}}$ costs a factor $(1 - L/W z_{\mathrm{CoM}})^{-1} \le 1.11$. (Along the measured trajectory, Sec. 0.1.3, the bound is unconditional.) (c) *Window length.* Since $\delta\phi(\tau) = (C_1/C_2)(\sinh C_2\tau - C_2\tau)$ is strictly monotone ($d\delta\phi/d\tau = \omega_{\mathrm{nom}} > 0$), the angle threshold determines τ_{end} bijectively: $C_2\tau_{\mathrm{end}} \in [2.1, 3.9]$ over the candidate rates, hence $\coth(C_2\tau_{\mathrm{end}}/2) \in [1.04, 1.28]$ and the bound is effectively time-free, $|\delta\omega/\omega| \le 1.3\,C_2\rho_{\max}/\dot M$.

Design consequence: the excitation cutoff and rate set. Because the propagated deviation is a computable function of the ramp rate and the excursion cap alone, the ideal analysis *selects* the excitation design. The systematic deviation grows with the cap and is worst at the slowest rate, crossing 1% at 4.8° for 0.10 N m/s and only at 8.9° for 1.20 N m/s; conversely the noise-limited residual floor falls with the cap (13% at 3°, 8% at 5°) and the post-onset sample count exceeds 40 everywhere. The 5° attitude cutoff is therefore the largest common cap that keeps the coherent deviation at the 1% level for every candidate rate, the binding constraint being the slowest ramp. In absolute terms the propagated deviation is 3.1–5.2 mrad/s at the 5° cap and at most 23 mrad/s at 9–10° — an order of magnitude below the in-motion gyro noise ($\approx$ 35 mrad/s RMS).

0.1.3 Transfer to the experimental configuration

The experiment departs from the ideal configuration in one respect: the sloped test surface places the rest attitude — and hence the onset, since the vehicle does not rotate before it — at $\phi^* \leq 2.6°$ (median 1.7°, measured over all 140 runs). The correct operating point is $(0, \phi^*, T^*)$, the expansion variable becomes the excursion $\delta\phi = \phi - \phi^*$ (measured: at most 9.4° over the fit window, median 6.1°), and every operating-point effect transfers with an explicit bound:

- the effective anti-restoring gradient shifts from $W z_{\text{CoM}}$ to $W z_{\text{CoM}} \cos\phi^* + W(l_p{+}\lambda_{\text{off}}) \sin\phi^*$ — at most +2.3%, absorbed by the calibrated C_2;
- the onset-instant small-angle bias grows from zero to $\frac{1}{2}W(l_p{+}\lambda_{\text{off}})\phi^{*2} \leq 5.4$ mN m per direction — common to both tip directions, which start from the same rest attitude, hence cancelling in M_{ff};
- the Lagrange remainder is in fact *smaller* at the tilted point (18.0 vs. 19.2 mN m at the 5° excursion), the restoring term subtracting from the curvature; the design tables, functions of the excursion alone, apply unchanged.

The slope's genuine effect is geometric, not modelling: it renders the two pitch directions asymmetric in runway ($\approx$ 3° vs. 7° to the absolute cutoff), lowering the SNR of the negative-pitch runs — precisely the regime for which the physics-constrained detector is needed.

On the experimental side every component of ρ is also *measured*: the recorded attitude and rotor speeds are evaluated through the exact expressions and projected out of the estimator span $\{1, \tau, \delta\phi\}$ — an algebraic identity, these being the estimator's exact per-run degrees of freedom. Table 1 compares the ideal-theory bounds with the measured values; the measured residuals, on which the claims rest, lie consistently below their a priori counterparts. For the ground effect the projection is evaluated with and without rotor–rotor interference: the measured slope ratio is +0.99% of $\dot{M}$ (interference neglected) and +4.20% (interference included, IQR $[4.13, 4.27]$%), *both* rate-invariant across the twelvefold rate range — the property the K-calibration absorption requires — and the out-of-span residual grows from 0.16 / 2.3 to 0.67 / 9.3 mN m (median RMS / max) with interference, while remaining edge-concentrated, so the propagated shape deviation stays at the few-percent level, below the noise-dominated residual. One term of the affine decomposition deserves explicit accounting. Because the moment coefficient is itself attitude dependent, the product $b(\phi)M_{\text{cmd}}$ contains $\beta_M\,\dot{M}\tau\,\delta\phi$ — a bilinear term in two spanned coordinates, hence the one piece of the ground-effect moment that is *not* in $\{1, \tau, \delta\phi\}$. Evaluated along the measured trajectories with $\beta_M = 0.0345$ rad^{-1} (`analysis/ge_bilinear.py`), its raw peak is 3.0 mN m and what survives the projection is 0.17 mN m at most (median 0.04) — the product is strongly correlated with the spanned coordinates along real trajectories, so it is absorbed almost entirely. The channel fates are model-independent: the thrust channel (rotor-common and body lift alike) acts through the arm l_p exactly, is antisymmetric between tip directions, and cancels in M_{ff} regardless of its magnitude; the moment coefficient leaves a relative bias on M_{ff} between −1.0% (no interference) and −4.3% (with interference), i.e. 3–14 mN m or 0.10–0.44 mm of CoM offset — below the validation scatter (52 mN m RMS), which is dominated by the genuine contact asymmetries the pivot-free average is designed to capture, and present identically at lift-off, where M_{ff} compensates it by design. The full chain of reductions is 85 → 28 → 12.3 → 1.2 mN m: the unconditional $\phi{=}0$ envelope at the largest absolute tilt (10°), the onset-anchored Lagrange bound at the median measured excursion, the measured pointwise maximum after projection, and the measured window RMS.

0.1.4 Empirical verification: a prediction test, not a fit

Because the constrained estimator has no free shape parameter per run — $C_1 = K\dot{M}$ with $\dot{M}$ measured, C_2 shared, baseline fixed by continuity — the post-onset curve is fully determined before the post-onset data are seen; goodness of fit is evidence, not flexibility. With NRMSE $= [\frac{1}{N}\sum_k(\omega_k - \hat{\omega}_k)^2]^{1/2} / \max_k |\omega_k - C|$, four measured, model-independent certificates close the claim:

Table 1: Components of the residual forcing ρ over the fit window. "A priori" bounds are the ideal-configuration values of Sec. 0.1.2 evaluated at the measured tilt range; "measured" values are evaluated along the recorded trajectories and projected out of the estimator span $\{1, \tau, \delta\phi\}$. All values in mN m.

Component	A priori	Measured (RMS / ma
Small-angle curvature	19.2 at 5° (≈ 28 at the median excursion)	1.2 / 12.3
Ground effect (without / with rotor interference)	3.2 / 7.5 (tangent deviation, full range)	0.16 / 2.3 – 0.67 / 9.3
Bilinear $\tau\,\delta\phi$ (GE moment channel)	3.9 (envelope of the factor maxima)	0.04 / 0.17
Ramp nonlinearity	30 (gate, full window)	1.6 / 6.2 (fit segment

1. *Rate invariance.* Across all 140 gate-passing runs the NRMSE is 6.3% (median; $R^2 = 0.95$) and flat from 0.10 to 1.20 N m/s (5.5–6.6%): a state-dependent deformation would grow with amplitude, and none does. The residual sits at 2.2× the quiet pre-onset gyro floor — motion- and thrust-induced measurement noise — while the predicted systematic deviation ($\leq 1\%$) lies below it.

2. *Trajectory-evaluated forcing.* The small-angle forcing deviates from the absorbable span by at most 12.3 mN m (median RMS 1.2), edge-concentrated where the kernel provides no amplification; the ground-effect forcing by 2.3 to 9.3 mN m depending on whether rotor interference is included. The measured slope ratios, +0.99% of $\dot{M}$ (without interference) and +4.20% (with), reproduce each model's coefficient in situ and are rate-independent in both cases — the property on which the K-calibration absorption rests.

3. *Window ablation.* Truncating the window at a 3° relative excursion — where the small-angle model is unimpeachable — changes the per-direction critical moments by ≤ 0.022 N m (within scatter) and M_{ff} by ≤ 0.012 N m (0.39 mm of CoM offset).

4. *Family exclusivity.* The $d \to 0$ quadratic limit, fitted over the same windows, incurs 14–75% truncation error: the data select the cosh member, not merely any smooth monotone curve.

In summary: structurally, every modeled effect renames the effective constants (C_2, K) that the calibration estimates, so to first order the response cannot leave the cosh family; the sole escape channel is measured — not merely bounded — at a few mN m and propagates to a sub-1%, sub-noise shape deviation. Empirically, the zero-freedom family predicts the response across all runs with a noise-dominated, rate-invariant residual, is insensitive to the window extent, and decisively outperforms its own quadratic limit. The theoretical bounds serve as conservative a priori envelopes; every headline figure is a measurement.